

MDS-UPDRS Finger Tapping Score

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested.

Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [87, 80, 87, 83, 89, 88, 88, 92, 93, 100]

Tap amplitudes (pixels): [177.28226081590904, 172.90459797240788, 160.52725625263767, 174.64249196572982, 177.5894140989265, 187.77912557044246, 180.06943105369106, 168.80758276807353, 166.20770138594662, 173.67786272291585, 176.13914953808538]

Gemini 3 Pro

To score the subject on MDS-UPDRS Item 3.4: FINGER TAPPING, we need to evaluate the provided features against the scoring criteria: speed, amplitude, hesitations, halts, and decrementing amplitude.

Analysis of Extracted Features:

1.

Rhythm and Interruptions (Hesitations/Halts):

- **Tap Periods:** [87, 80, 87, 83, 89, 88, 88, 92, 93, 100]
- The periods are relatively consistent, ranging mostly between 80 and 93 frames.
- There is a slight increase in period length towards the end (92, 93, 100), indicating a very minor slowing, but there are no sudden spikes that would suggest a hesitation, interruption, or freeze (arrest). The rhythm appears stable.

2.

Speed (Slowing):

- The periods are fairly long (around 80-100 frames). Without the frame rate, it's hard to judge absolute speed, but the consistency suggests no significant *pathological* slowing or fatigue-induced slowing during the task, other than the minor drift at the very end.

3.

Amplitude (Decrementing):

- **Tap Amplitudes:** [177, 173, 161, 175, 178, 188, 180, 169, 166, 174, 176] (rounded)
- The amplitude starts around 177.
- It fluctuates slightly but remains robust throughout the sequence.

- There is no consistent decrement. For example, tap 6 (188) is higher than tap 1 (177). The final taps (174, 176) are very close to the starting amplitude.
- There is no evidence of decrementing amplitude near the end, midway, or after the first tap.

Scoring Conclusion:

- **Interruptions:** None observed (0).
- **Slowing:** No significant slowing observed (0).
- **Decrementing Amplitude:** No decrement observed (0).

The subject performs the task with a stable rhythm and consistent amplitude without interruptions or decrements.

Score: 0
